Fiscal Year:	FY 2014	**Task Last Updated:**	FY 12/23/2014
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Sensitivity Modulation of Different T-Cells Activation Pathways by Microgravity (PADIAC)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	milliehf1@gmail.com	**Fax:**	FY 415-750-6667
PI Organization Type:	NON-PROFIT	**Phone:**	415-750-6940
Organization Name:	Northern California Institute for Research and Education		
PI Address 1:	Lab of Cell Growth, Mail Code 151F, VAMC		
PI Address 2:	4150 Clement Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	ESA-RA-LS-01-LSRA
Start Date:	07/01/2007	**End Date:**	12/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Soyuz/ISS		
Flight Assignment:	ISS NOTE: New end date is 12/30/2013 per A. Chu/ARC and NSSC (Ed., 2/21/2013) NOTE: New end date is 6/30/2013 per A. Chu/ARC (Ed., 1/27/2012) NOTE: to be extended to 6/30/2013 per A. Chu/ARC (Ed., 1/17/2012) NOTE: Extended to 6/30/2012 (from 6/30/2010) per A. Chu/ARC (6/2010)		
Key Personnel Changes/Previous PI:	PI is Dr. Isabelle Walther of Switzerland. Dr. Hughes-Fulford is the US PI (ESA CoI).		
COI Name (Institution):	Walther, Isabelle (Swiss Federal Institute of Technology)		
Grant/Contract No.:	NNX07AM98G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This is an international collaboration between Dr. Millie Hughes-Fulford, Dr. Isabelle Walther, Augusto Cogoli, Proto Pippia, Mariantonia Meli, Grazia Galleri. Mammalian cells subjected to conditions of simulated microgravity on ground as well as to space flight conditions are showing alternations in their structure and function. Changes in proliferation, differentiation, and genetic expression have been demonstrated in several types of cells (reviewed in Cogoli 1996, Cogoli and Greuter 1997). Among the mammalian cells, the cells of the immune system, in particular, are severely affected by the space environment. Especially T-lymphocytes are altered in their activation process. Change in growth rate, cytokine production, locomotion, gene expression, and PKC distribution have been observed on Space Shuttle flights (Tipton et al. 1996, Pellis et al. 1997, Cogoli-Greuter et al. 1997, Hatton et al. 1999, Lewis et al. 2001). At the beginning of the space research, the experiments are based on simple question with answers like “yes/no”; at the present time the knowledge on the signal transduction mechanisms are progressing rapidly and the research is orienting to answer deeper questions about regulation pathways and to get a better understanding at the molecular level. Moreover, the laboratory techniques have evolved greatly. Sub-populations of T-cells can be isolated from peripheral blood, the expression of specific genes can be measured, and signal pathways can be specifically activated. This allows studying more complex and subtle reactions in cells biology. The goal of this proposal is to study, thanks to these new methods, the finer regulation of lymphocyte activation, especially at the level of the cytokines IL-2, IL-1, and their receptors using different activation pathways. The specific objectives are to further investigate, using several activators, the genetic expression of the co-stimulatory transduction pathway mediated by the CD28, to assess whether the IL-1 receptor is involved in the inhibition of the activation, to see if the activation is also restored in space as it is in simulated microgravity using anti-CD3/IL-2, and to determine which genes are then expressed under these special conditions. Microgravity will be used as an inhibitor of activation without having to use chemicals putatively damaging the cells. The main hypothesis to be tested is whether the restoring of the activation under the specific anti-CD3/IL-2 activation is due to a restoring of the receptor (IL-2R and/or IL-1R) gene expression. Correlated aspects such as the expression of early oncogenes and the production cytokines and their receptor will also be studies. The experiment will be based on an on-going ground-based research using the RPM as the main tool.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Immune response is a key element in continued health. It is known that there is a suppression of the the immune system during spaceflight. Our studies are aimed at understanding the molecular defect of the T cell that suppresses activation during microgravity. Once we understand the controlling pathways, by using microgravity as an investigative tool, we can identify pharmaceutical targets of the immune system. These targets will include those that enhance T cell function as well as targets to blunt immune function to treat autoimmune diseases such as lupus and Rheumatoid arthritis.
Task Progress:	(Ed. note: compiled from PI's final report) The flight experiment was completed in 2011 and we found differences in immune response using either ConA/antiCD28, magnetic beads CD3/CD28 and CD3, or IL-2 activation methods. We found that both activation mechanisms of ConA/CD28 and CD3/CD28 activate the immune system. We also found that the absence of CD28 could replace induction of itself (IL2) and IL-2 could replace the need for CD28. We found that microgravity caused severe reduction of key immune gene expression of IL2, IL2ra, TNFa, Xcl2, EGR1, CCL20, GMCSF, and IFNg.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Grenon SM, Jeanne M, Aguado-Zuniga J, Conte MS, Hughes-Fulford M. "Effects of gravitational mechanical unloading in endothelial cells: association between caveolins, inflammation and adhesion molecules." Scientific Reports. 2013;3:1494. http://dx.doi.org/10.1038/srep01494 ; PubMed PMID: 23511048; PubMed Central PMCID: PMC3603133 , Mar-2013
Articles in Peer-reviewed Journals	Chang TT, Hughes-Fulford M. "Molecular mechanisms underlying the enhanced functions of three-dimensional hepatocyte aggregates." Biomaterials. 2014 Feb;35(7):2162-71. Epub 2013 Dec 12. http://dx.doi.org/10.1016/j.biomaterials.2013.11.063 ; PubMed PMID: 24332390; PubMed Central PMCID: PMC4012541 (Available on 2015/2/1) , Feb-2014
Articles in Peer-reviewed Journals	Grenon SM, Owens CD, Alley H, Chong K, Yen PK, Harris W, Hughes-Fulford M, Conte MS. "n-3 Polyunsaturated fatty acids supplementation in peripheral artery disease: the OMEGA-PAD trial." Vasc Med. 2013 Oct;18(5):263-74. http://dx.doi.org/10.1177/1358863X13503695 ; PubMed PMID: 24052491 ; PubMed Central PMCID: PMC4074651 , Oct-2013